

CASE CHALLENGE

"LEG ULCERS"

SURGERY SHORTCASES

Dr.Kishan Rao



CASE PRESENTATION

By Mr. Melwin Maben

4th Year MBBS, HIMS Hassan.

Name: Mr XYZ

Age: 50 years

Sex: Male

Address: Holenarsipura, Hassan

Education: 7TH Std.

Occupation: Farmer

Socioeconomic status: Class 4 of Modified BG Prasad Classification

DOA: 7th January 2020

DOE: 14th January 2020

CHIEF COMPLAINTS

C/o Ulcer over Right foot since 3 weeks

C/o Discharge from the ulcer since 3 weeks

HISTORY OF PRESENTING ILLNESS

Patient is a known case of Diabetes Mellitus since 16 years and on regular treatment, sustained trauma due to a stone 1 month back which formed a swelling on the outer aspect of right foot. The swelling then burst developed a ulcer 3 weeks back on the outer aspect of right foot which was initially of approximate size of 2x2 cm but has now progressed to an approximate size of 5x5 cm.

He also complains of serosanguinous foul smelling discharge from the ulcer since 3 weeks which was moderate in quantity.

No H/o pain in the region of ulcer

No H/o altered sensation

PAST HISTORY

No H/o any similar complaints in the past

No H/o Hypertension, TB, epilepsy or asthma

No H/o hospitalization or surgeries in the past

FAMILY HISTORY

No H/o any similar complaints in the family members

No H/o Diabetes Mellitus in the family members

PERSONAL HISTORY

- Diet: Mixed
- Appetite: Normal
- Sleep: Normal
- Bowel and Bladder Movements: Normal
- No H/o any addictions
- No H/o any known allergies

SUMMARY

A 50 year old gentleman who is a known case of Diabetes Mellitus since 16 years on regular medications presents with ulcer on outer aspect of right foot since 3 weeks after sustaining a minor trauma 1 month back. The ulcer is associated with serosanguinous, foul smelling discharge since 3 weeks.

PHYSICAL EXAMINATION

- Here is a middle aged gentleman who is moderately built and nourished. He is conscious, cooperative and well oriented to time, place and person.
- Height: 170 cm
- Weight: 70 kgs
- BMI: 24.22 kg/m_2
- Pallor, Icterus, Clubbing, Cyanosis, Lymphadenopathy, Edema: Not present

VITALS

- Pulse rate – 86 beats/min, regular rhythm, normal volume, and no radio-radial or radio-femoral delay
- Blood pressure – 110/70 mmHg. Taken from right hand in sitting position
- Respiratory rate – 20 breaths/min
- Temperature – 98.6 F

LOCAL EXAMINATION

INSPECTION

- A solitary ulcer present on the medial aspect of right foot below the lateral malleolus extending upto the plantar aspect which measures approximately 7x 5 cm, irregular in shape, the edges are sloping, edematous and inflammed. The floor is covered with granulation tissue and slough.
- There is seropurulent discharge which is foul smelling.
- The surrounding area is inflammed and slightly pigmented
- The rest of the right lower limb is normal
- All joint movements are normal

PALPATION

- no local rise of temperature or tenderness
- solitary ulcer present on the right foot, below the lateral malleolus extending up to plantar aspect, measuring 7x5 cm in size, irregular in shape. There is a sloping edge with distinct margin. The depth is uneven varying from 0.5 to 1 cm. The base is made of subcutaneous tissue.
- no induration and the ulcer is mobile
- ulcer does not bleed on touch
- surrounding skin is inflammed and slightly pigmented.
- no loss of sensation in other parts of the right lower limb.

EXAMINATION OF LYMPH NODES

no enlarged lymph nodes in the inguinal region.

EXAMINATION OF VACULAR INSUFFICIENCY

no signs suggestive of vascular insufficiency

EXAMINATION OF NERVE LESIONS

no signs suggestive of nerve lesions

SYSTEMIC EXAMINATION

CARDIOVASCULAR SYSTEM

S1 and S2 heard, No murmurs

RESPIRATORY SYSTEM

- Trachea: Central
- Bilateral equal air entry present
- Vesicular Sounds are heard, No added sounds

PER ABDOMEN

Soft, tender and no organomegaly

CENTRAL NERVOUS SYSTEM

No focal deficit present

SUMMARY

A 50 year old gentleman who is a known case of Diabetes Mellitus since 16 years on regular medications presents with ulcer on inner aspect of right foot since 3 weeks after sustaining a minor trauma 1 month back. The ulcer is associated with serosanguinous, foul smelling discharge since 3 weeks.

On inspection an ulcer is seen on the right lower foot extending from below the lateral maleolus to the plantar aspect, irregular in shape, with sloping edges and floor is covered with granulation tissue and slough. There is slight seropurulent, foul smelling discharge.

All inspectory findings are confirmed on palpation. The base is not induarted and the ulcer is mobile. It does not bleed on touch.

PROVISIONAL DIAGNOSIS

A 50 year old gentleman who is a known case of Diabetes Mellitus with chronic non-healing ulcer on lateral and plantar aspect of right foot

INVESTIGATIONS

1. Routine examination of the blood
2. Examination of the Urine
3. Bacteriological examination of the discharge of the ulcer
4. Skin test
5. Chest X-ray
6. Biopsy
7. X-ray of the Bone and Joint
8. Contrast Radiography
9. Doppler

MANAGEMENT

-Control of blood sugar levels

-Elevation of the limb and bed rest

-Avoiding any trauma to the limb

-Debridement of ulcer

-Antibiotics

-Regular cleaning and dressing of the wound

DEFINITIONS

- **Wound**: break in the integrity of the skin or tissues often, which may be associated with disruption of the structure and function.



- **Ulcer**: break in the continuity of the covering epithelium, either skin or mucous membrane due to molecular death



- **Necrosis**: localised area of death of tissue followed by degradation of tissue by hydrolytic enzymes liberated from dead cells
- **Gangrene**: Gangrene is a form of necrosis of tissue with superadded putrefaction







- **<u>Granulation Tissue</u>**: proliferation of new capillaries and fibroblasts intermingled with red blood cells and white blood cells with thin fibrin cover over it.



- **<u>Eschar</u>**: necrotic tissue found in a full-thickness wound



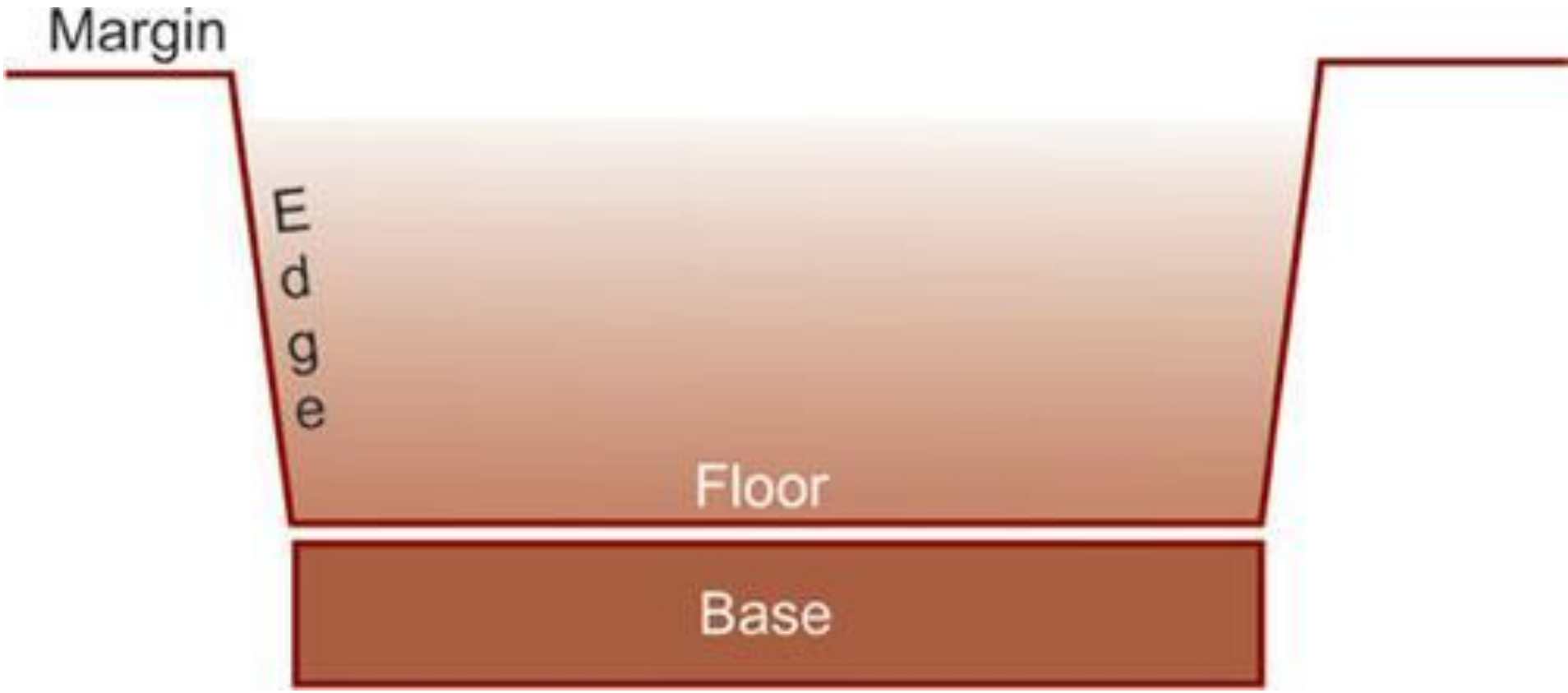


- **Cellulitis**: spreading inflammation of subcutaneous and fascial planes.
- **Induration**: clinical palpatory sign which means a specific type of hardness in the diseased tissue.

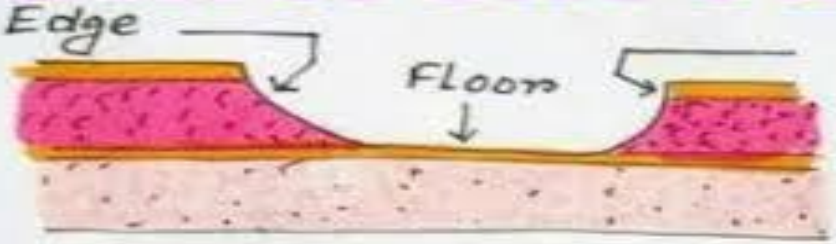


CELLULITIS OF LEG

ULCER



Different edges of Ulcer





Undermined edge → Tuberculous ulcer



Punched out edge → Trophic ulcer (Diabetic ulcer)



Sloping edge → Healing ulcer

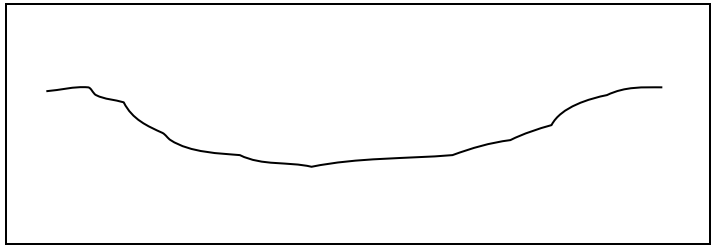


Everted → Squamous cell carcinoma



Rolled → Basal cell carcinoma

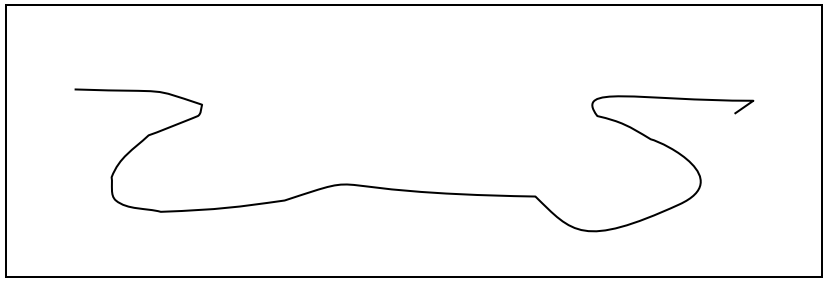
SLOPING EDGE





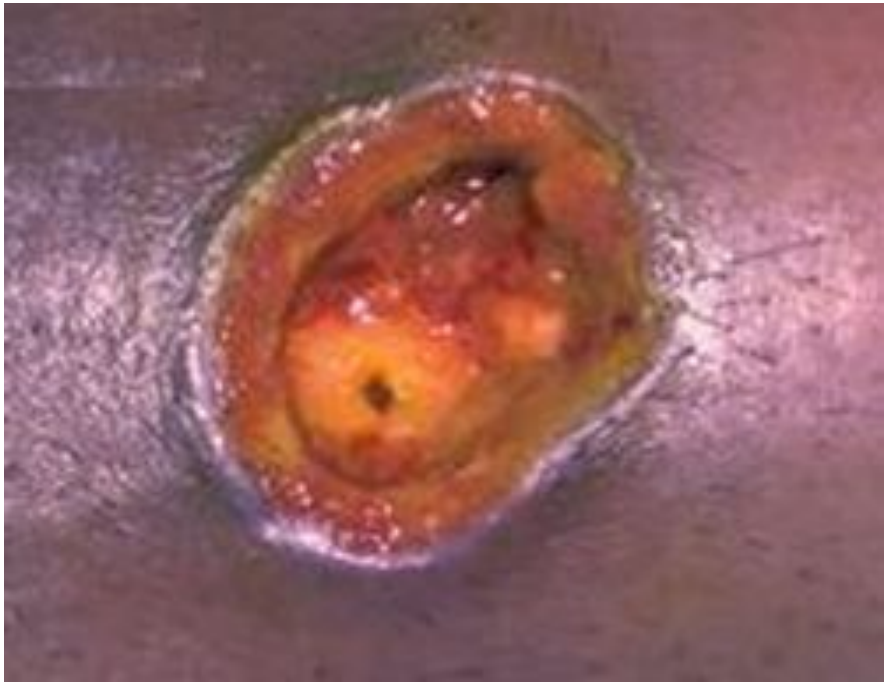


UNDERMINED EDGE









Tubercular ulcer

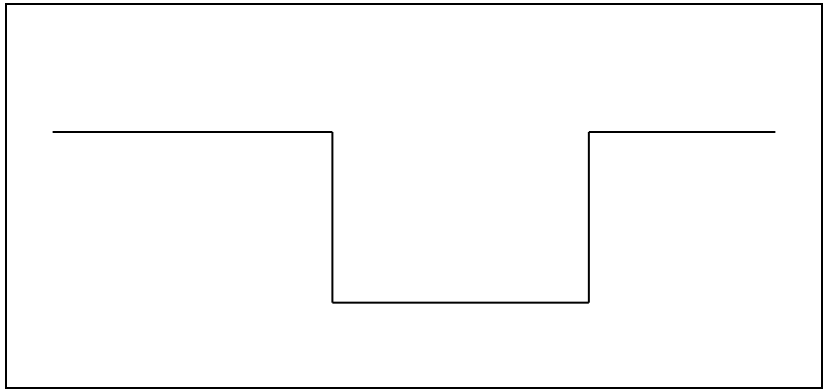


Diabetic foot infection



Pressure bed sore in the buttock. Subcutaneous fat is more susceptible to pressure than the skin

PUNCHED OUT EDGE







Syphilis

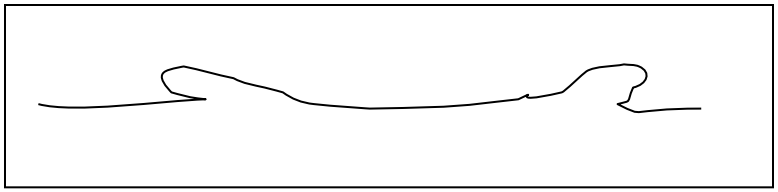


Leprosy



Neuropathic ulcer in the sole of foot

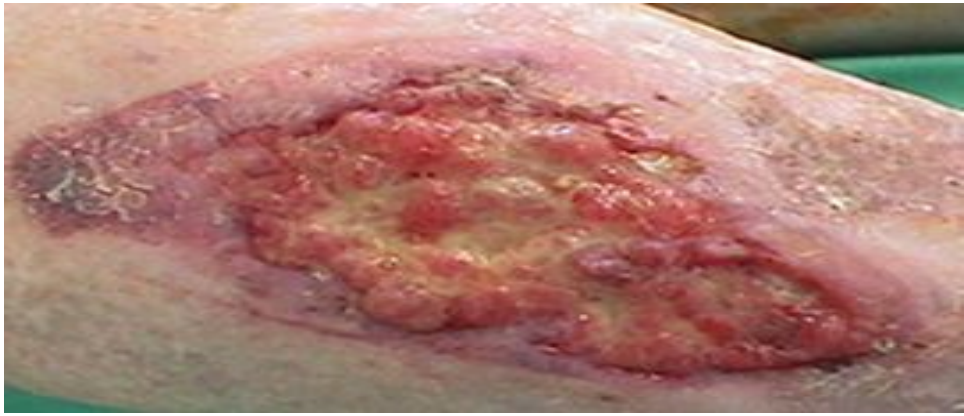
EVERTED EDGE





SQUAMOUS CELL CARCINOMA

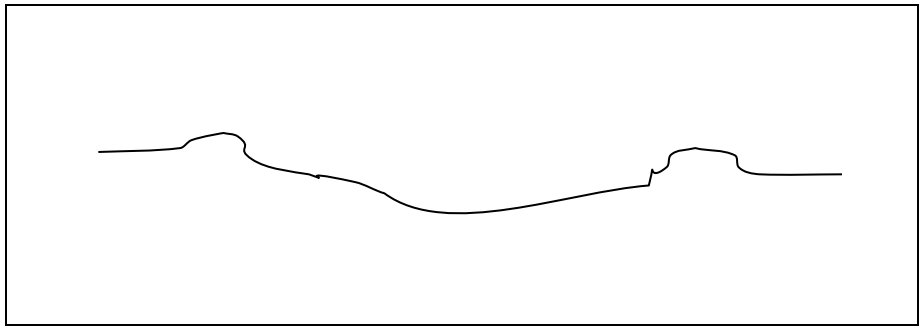




Malignant transformation in a chronic venous ulcer

"Marjolin" ulcer

RAISED AND BEADED





BASAL CELL CARCINOMA





Floor

Granulation tissue

Slough

Gangrenous tissue

Deeper tissues as tendon or bone











Discharge

serous, sanginous, serosanguinous , purulent

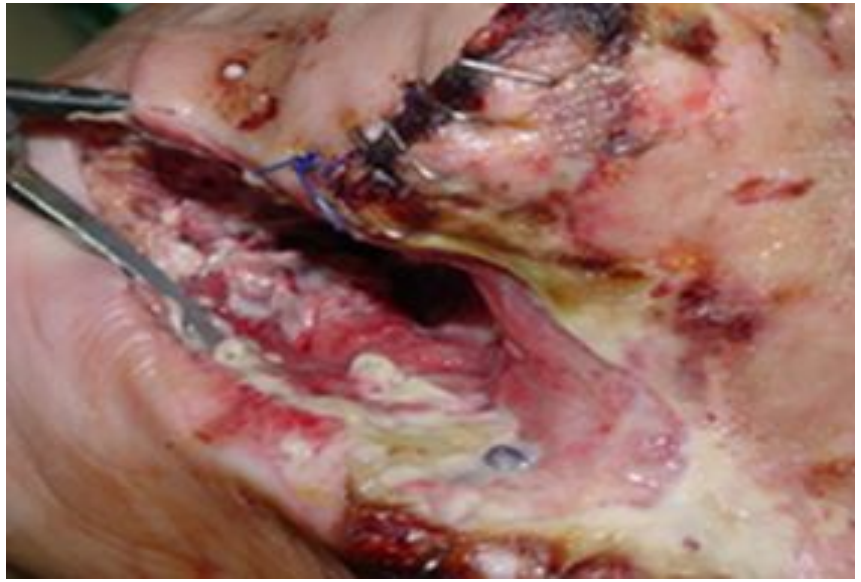
A dry discharge forms a scab



Purulent discharge from infected bed sore



Dry scab covering the ulcer



Serosanguinous discharge from infected diabetic foot ulcer

Surrounding Skin



Pigmentation around venous ulcer



Cellulitis around the ulcer



Hyperpigmentation of recurrent infection

TYPES OF ULCERS

TRAUMATIC ULCERS



DENTAL ULCER



MARJOLIN'S ULCER

ARTERIAL ULCER



BUERGER'S DISEASE

VENOUS ULCERS



Fig. 1: Venous ulcer before treatment and 1 month after



TROPHIC ULCERS



LEPROSY ULCER



BED SORE

DIABETIC ULCER





TUBERCULOUS ULCER



LUPUS VULGARIS



SYPHILITIC ULCER



HARD CHANCRE



GUMMA

MELENEY'S ULCER



Figure 1 | Two ulcers were present at the medial border of the right hallux and one ulcer was found under the nail plate of the medial fold of the right hallux

MALIGNANT ULCERS

Basal cell carcinoma



Squamous cell carcinoma



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MALIGNANT MELANOMA



BAZIN'S ULCER



MARTOREL'S ULCER



CLINICAL ASSESSMENT OF LEG ULCER

FEATURES	VENOUS	ARTERIAL	NEUROTROPHIC	TRAUMATIC	MALIGNANT
SITE	gaiter area 70%, lateral 20%, circumferential 5%	malleoli, heel, metatarsal heads, 5 MT base	pressure area	site of trauma	face, lips, tongue
SIZE/SHAPE	large shallow , vertically oval	small deep	variable	variable	variable
MARGIN	irregular	regular	regular		
EDGE	sloping		punched out	sloping	everted/rolled
FLOOR	granulation	slough/ necrotic	slough	variable	black mass











FEATURES	VENOUS	ARTERIAL	NEURO	TRAUMA	MALIGNANT
BASE	fix /no induration	no	no	no	indurated
DEPTH	shallow	deep	deep up to bone	variable	shallow
DISCHARGE	serous/ purulent/ sero-sanguinous / greenish				
TENDER	painless	painful	painless	painful	painless
SKIN	edema/eczema/ pigmentation/ blanched/ varicose vein	Thin skin/ decreased hair growth/ loss of sc fat/ loss of luster/ brittle nails	callus/ loss of sensation	cellulitis	fix to underlying structures
LYPMH NODE	absent	absent	absent		present except BCC






Leg Ulcer
Clinical Case Proforma

- **History**

1.Duration
2.Mode of onset
3.Site
4.Progression
5.Pain over ulcer
6.Discharge
7.Associated disease
8.History of similar ulcer in past, TB
9.Personal History

- **Physical Examination**
  - General Survey
  - Systemic examination

Local Examination

Inspection

1.Site
2.Shape
3.Size
4.Number
5.Extent
6.Margin
7.Edge
8.Floor
9. Discharge
10.Adjacent area

Palpation

1.Temperature
2.Tenderness
3.Size of ulcer
4.Margin
5.Edge
6.Floor
7.Base
8.Mobility
9.Discharge
10.Surrounding area

Examination of

- Regional lymph nodes
- Adjacent joints
- Vascular
- Neurological

THANK YOU

